Statistics

Academic Year: (2026 / 2027) **Review date: 30/04/2026 11:04:44**

Department assigned to the subject: Statistics Department
Coordinating teacher: CASCOS FERNANDEZ, IGNACIO
Type: Basic Core ECTS Credits : 6.0
Year : 1 Semester : 1
Branch of knowledge: Social Sciences and Law

LEARNING OUTCOMES

CB1: Students have demonstrated possession and understanding of knowledge in an area of study that builds on the foundation of general secondary education, and is usually at a level that, while relying on advanced textbooks, also includes some aspects that involve knowledge from the cutting edge of their field of study.
CE.FB1: Ability to solve mathematical problems that may arise in engineering. Ability to apply knowledge of: linear algebra; geometry; differential geometry; differential and integral calculus; differential and partial differential equations; numerical methods; numerical algorithms; statistics and optimisation.
RA1: Have basic knowledge and understanding of mathematics, basic sciences, and engineering within the aerospace field, including: behaviour of structures; thermodynamic cycles and fluid mechanics; the air navigation system, air traffic, and coordination with other means of transport; aerodynamic forces; flight dynamics; materials for aerospace use; manufacturing processes; airport infrastructures and buildings. In addition to a specific knowledge and understanding of the specific aircraft and aero-engine technologies in each of the subjects included in this degree.

OBJECTIVES

Once successfully having studied this subject, the students should be able to:

- Analize problems involving random phenomena
- Define populations for a statistical study
- Build Hypothesis about a distribution
- Test hypothesis about the paramters of the chosen model
- Evaluate how well does the model fit to reality
- Understand the limitations of the methods that have been studied and the conditions under which they lead to wrong conclusions

DESCRIPTION OF CONTENTS: PROGRAMME

BLOCK 0: DESCRIPTIVE STATISTICS
0. Descriptive Statistics
BLOCK I: PROBABILITY
1. Introduction to Probability
1.1 Introduction
1.2 Random phenomena
1.3 Definition of probability and properties
1.4 Assessment of probabilities in practice
1.5 Conditional probability
1.6 Bayes Theorem
2. Random variables
2.1 Definition of random variable
2.2 Discrete random variables
2.3 Continuous random variables
2.4 Characteristic features of a random variable

2.5 Transformations of random variables
2.6 Independence of random variables
BLOCK II: PARAMETRIC MODELS AND INFERENCE
3. Distribution models
3.1 Binomial distribution
3.2 Geometric distribution
3.3 Poisson distribution
3.4 Uniform distribution (continuous)
3.5 Exponential distribution
3.6 Normal distribution (with CLT)
4. Statistical Inference
4.1 Introduction
4.2 Estimators and their distributions
4.3 Confidence Intervals
4.4 Hypothesis testing
4.5 Particualr tests on a single sample
4.6 Comparison of two populations
BLOCK III: APPLICATIONS
5. Quality control
5.1 Introduction, control charts
5.2 Variables control charts, the X-bar chart
5.3 Attributes control charts, the p and np charts
6. Linear regression
6.1 Introduction
6.2 Simple linear regression
6.3 Multiple linear regression

LEARNING ACTIVITIES AND METHODOLOGY

The Learning Activities are (approximate hours for each are indicated in parentheses)
AF1:Theoretical-Practical Sessions (44 hours)
AF2: Workshops and Laboratories (6 hours)
AF3: Tutorials (4 hours)
AF4: Independent or Group Work by the Students (98 hours)
AF9: Final Exam (4 hours)

The Teaching Methodologies are
MD1: Lectures
MD2: Practical Sessions
MD3: Laboratory Sessions
MD4: Tutorials

ASSESSMENT SYSTEM

% end-of-term-examination/test:	0
% of continuous assessment (assigments, laboratory, practicals…):	100

There will be continuous evaluation by means of two partial examinations (45%+55%). At the partial examinations there will be some questions about the computer sessions.

If the grade obtained at the continuous evaluation is 6 or higher (has followed the continuous evaluation satisfactorily), the student should not attend the final exam and his/her final grade will be the grade of the cotinuous evaluation.

If the grade obtained at the continuous evaluation is lower than 6, the student will have to attend the final exam. For those students, the final grade will be computed giving a 40% weight to the partial examinations, and a 60% weight to the grade at the final exam.

BASIC BIBLIOGRAPHY

- MONTGOMERY, D.C., RUNGER, G.C. Applied Statistics and Probability for Engineers, John Wiley & Sons, 2003

- Navidi, W. Statistics for Engineers and Scientists, McGraw-Hill, 2006

ADDITIONAL BIBLIOGRAPHY

- GUTTMAN, L., WILKS, S.S., HUNTER, J.S. Introductory Engineering Statistics, Wiley, 1992.